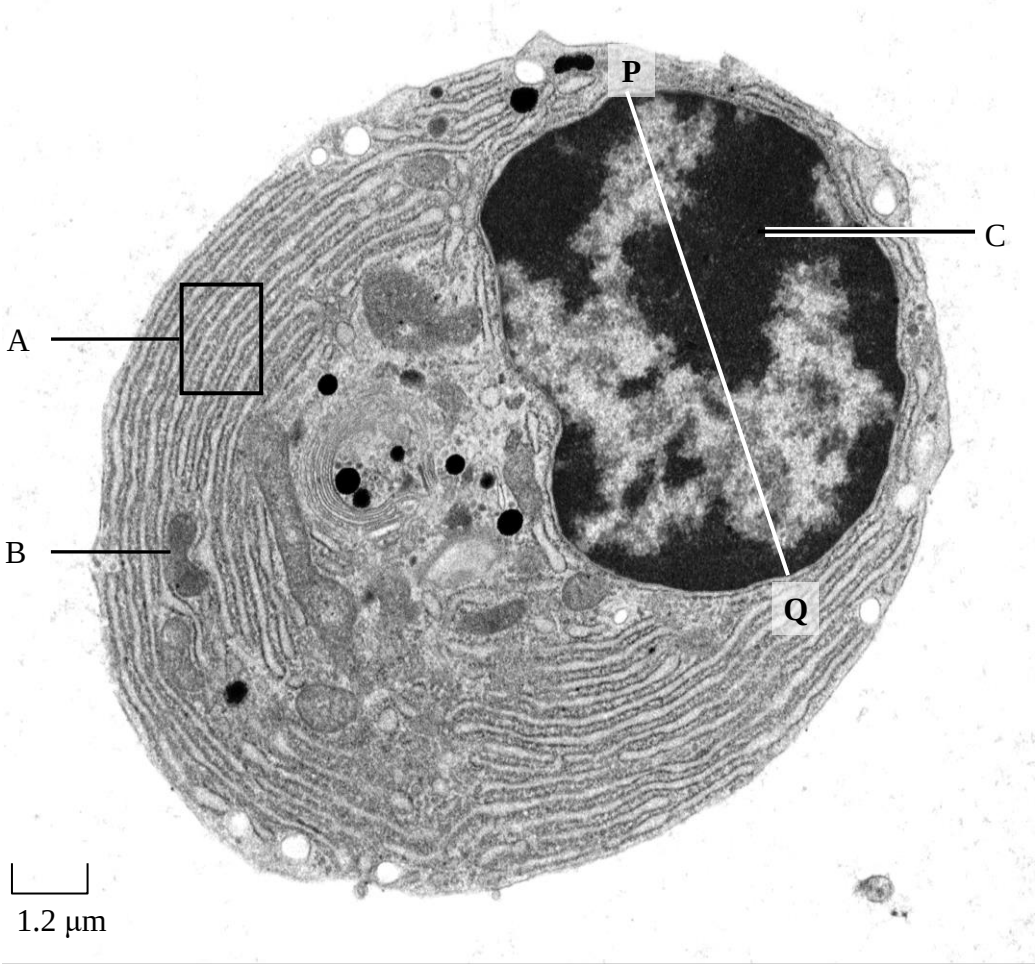
Short questions

|!|BCESB0030000001|!| ★

The following electron micrograph shows a type of human cell.



(a) Identify the structures labelled as A, B and C. (3 marks)

A: _____

B: _____

C: _____

(b) Calculate the magnification of the image. (1 mark)

_____

(c) Calculate the actual length of PQ. Show your working. (2 marks)

_____

_____

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(d) Give ***one*** advantage of using a light microscope over an electron microscope.

(1 mark)

_____

_____

##

(a) A: rough endoplasmic reticulum (1)

B: mitochondrion (1)

C: nucleus (1)

(b) 8333× (1)

(c) 6.7 × 1.2 μm (1)

= 8.04 μm (1)

(d) Any ***one*** of the following: (1)

- Light microscopes produce colour images instead of black and white as in electron microscopes.
- Light microscopes can be used to view living organisms.
- Sample preparation for light microscopes is easier.
- Light microscopes are cheaper / more portable.

(Or other reasonable answers)

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	3
(b)	Mathematical skills	1
(c)	Mathematical skills	2
(d)	Applying concepts	1

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|!|BCESB0030000002|!| ★

The photomicrograph below shows a type of single-celled organism.



(Credits: Picturepest, Wikimedia Commons,
https://commons.wikimedia.org/wiki/File:Ameba_002.jpg)

(a) Name structures Y and Z. (2 marks)

Y: _____

Z: _____

(b) What is the function of structure X? (1 mark)

_____

_____

(c) Explain why specimen containing this organism has to be stained before viewing with a light microscope. (1 mark)

_____

_____

(d) State ***one*** organelle present in plant cells but not in this organism. (1 mark)

_____

##

(a) Y: nucleus (1)
Z: cytoplasm (1)

(b) Structure X (cell membrane) controls the movement of substances into and out of the cell. (1)

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(c) Some parts of the cell would be stained. This provides contrast for easier observation. (1)

(d) Cell wall / chloroplast (1)

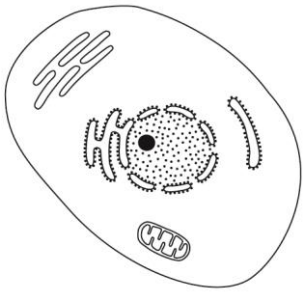
(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	1
(c)	Practical skills	1
(d)	Understanding basic concepts	1

|!|BCESB0030000003|!| ★

The diagram below is drawn based on an electron micrograph of an animal cell.



(a) A student wants to examine the animal cell using a light microscope at medium power magnification. Describe the steps which should be taken to set up a light microscope for viewing a slide under the medium power objective lens. (5 marks)

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(b) Sketch a diagram to show the appearance of the animal cell viewed at 200×.

(2 marks)

##

(a) Insert a low power eyepiece into the body tube. (1)

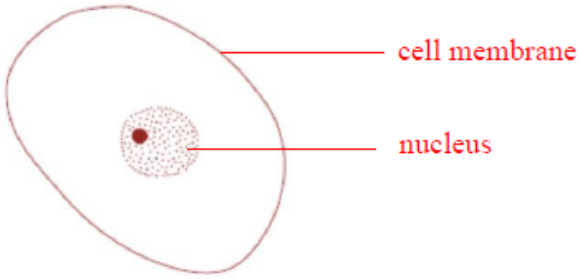
Switch on the light source / adjust the mirror to reflect light through the hole on the stage. (1)

Put the slide on the stage. Adjust the condenser and iris diaphragm until the slide is illuminated evenly. (1)

Choose a low power objective. Bring the image into focus using the coarse adjustment knob. (1)

Switch to a medium power objective. Bring the image into focus using the coarse adjustment knob and fine adjustment knob. (1)

(b)



Correct drawing (1)

Correct labels (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Practical skills	5
(b)	Presenting information	2

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|!|BCESC0030000004|!|

For each of the cell structures listed in column 1, select from column 2 certain phrase(s) that correctly describe(s) it. Put the appropriate letter(s) in the space provided. (5 marks)

Column 1		***Column 2***
Cell wall	(a) _____	A. described as fluid mosaic
Cell membrane	(b) _____	B. contains proteins
		C. made up of carbohydrates
		D. fully permeable
		E. differentially permeable

##

(a) C, D (1, 1)

(b) A, B, E (1, 1, 1)

_____

(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	3

|!|BCESB0030000005|!| ★ **Cross topic**

For each of the cell structures listed in column 1, select from column 2 one phrase that correctly describes it. Put the appropriate letter in the space provided. (4 marks)

Column 1		***Column 2***
nucleus	(a) _____	A. site of photosynthesis
ribosome	(b) _____	B. translation of mRNA
smooth ER	(c) _____	C. aerobic ATP production
mitochondrion	(d) _____	D. mRNA synthesis
		E. helps in synthesis and transport of proteins
		F. helps in synthesis and transport of lipids

##

(a) D (1)

(b) B (1)

(c) F (1)

(d) C (1)

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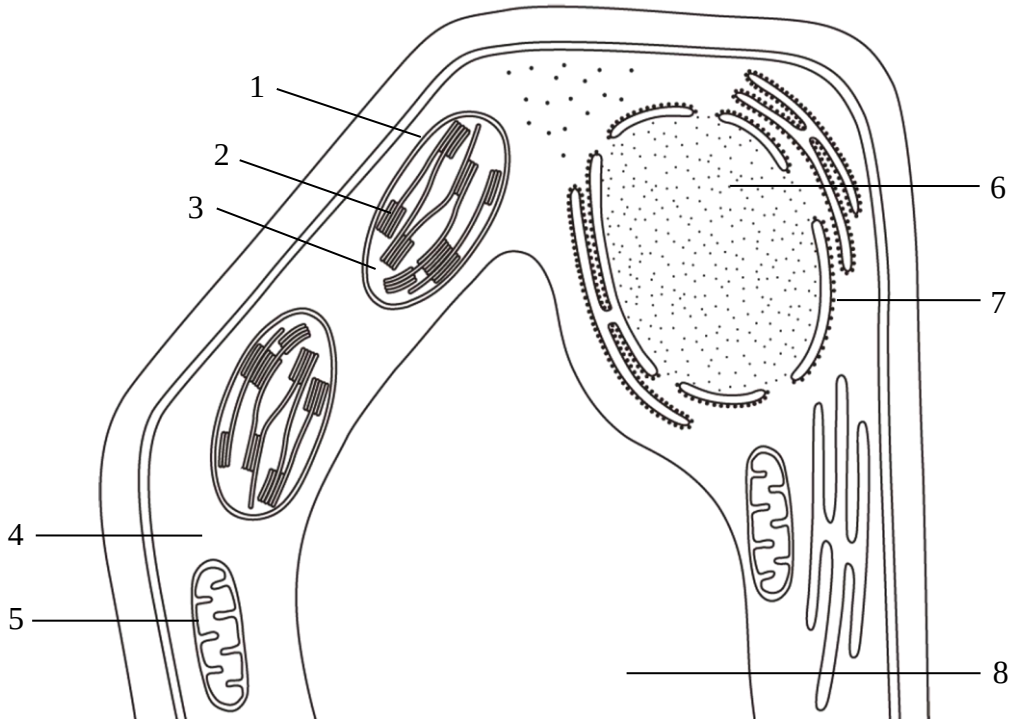
(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

|!|BCESB0030000006|!| ★ **Cross topic**

The diagram below shows the structure of part of a plant cell.



Using the numbers given in the diagram, indicate the site where each of the following processes takes place. (6 marks)

(a) CO_2 production ☐
(b) CO_2 fixation ☐
(c) glycolysis ☐
(d) photolysis of water ☐
(e) transcription ☐
(f) translation ☐

##

(a) 5 (1)
(b) 3 (1)
(c) 4 (1)

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(d) 2 (1)

(e) 6 (1)

(f) 7 (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1
(f)	Understanding basic concepts	1

|!|BCESB0030000007|!| ★

For each of the cell listed in column 1, select from column 2 one phrase that matches it. Put the appropriate letter in the space provided. (3 marks)

Column 1		***Column 2***
Leaf epidermal cell	(a) _____	A. contains chloroplasts
Bacterial cell	(b) _____	B. contains cellulose cell wall
Epithelial cell lining the trachea	(c) _____	C. contains plasmids
		D. contains cilia on cell surface
		E. contains microvilli

##

(a) B (1)

(b) C (1)

(c) D (1)

(3 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

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|!|BCESC0030000008|!|

For each of the cell structures listed in column 1, select from column 2 one function that matches it. Put the appropriate letter in the space provided. (3 marks)

Column 1		***Column 2***
Mitochondrion	(a) _____	A. for locomotion
Flagellum	(b) _____	B. site of lipid synthesis
Rough endoplasmic reticulum	(c) _____	C. site of protein synthesis
		D. controls cellular activities
		E. site of ATP synthesis

##

(a) E (1)

(b) A (1)

(c) C (1)

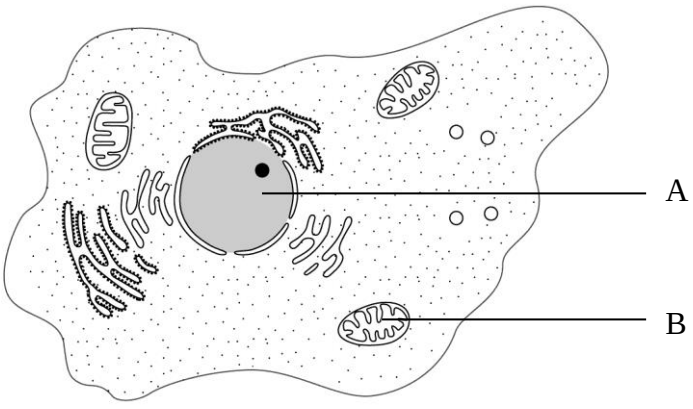
(3 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

|!|BCESB0030000009|!| ★

The following diagram shows the structure of an *amoeba*.



(a) Identify structure A and state its function. (2 marks)

_____

_____

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(b) Structures A and B are surrounded by a double membrane. Give ***two*** structural differences between the double membranes of structures A and B. (4 marks)

(c) Besides structures A and B, state ***one*** other structure observable in the diagram, which shows that *amoeba* is a eukaryote. (1 mark)

##

(a) Structure A is a nucleus. (1)

It houses the genetic material DNA / controls all cellular activities. (1)

(b) Any ***two sets*** of the following differences: (2, 2)

Structure A (nucleus)	**Structure B (mitochondrion)**
• inner membrane is not highly folded (1)	• inner membrane is highly folded (1)
• membrane has pores (1)	• membrane has no pores (1)
• outer membrane is continuous with the endoplasmic reticulum (1)	• outer membrane is not continuous with the endoplasmic reticulum (1)

(c) presence of endoplasmic reticulum (1)

(7 marks)

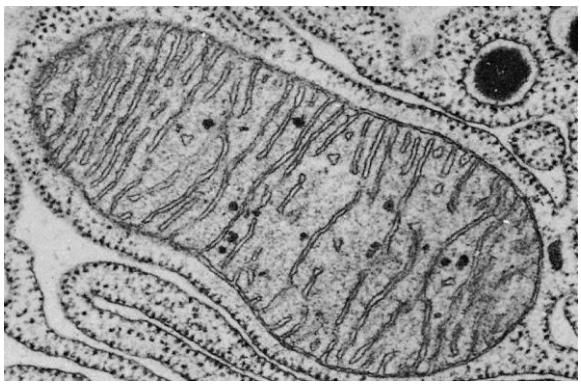
##

	Skills assessed	**Marks**
(a)	Analysing data	1
	Understanding basic concepts	1
(b)	Applying concepts	4
(c)	Understanding basic concepts	1

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|!|BCESB0030000010|!| ★

The micrograph below shows an organelle X.



(a) Name organelle X. (1 mark)

(b) Explain why organelles X are found abundantly in human liver cells. (2 marks)

(c) State another type of organelle which is abundant in liver cells. Relate its abundance to the function of liver cells. (2 marks)

(d) Suggest and explain another cell type which contains a lot of organelle X. (2 marks)

##

(a) X is a mitochondrion. (1)

(b) X is the site of respiration to produce ATP. (1)
A lot of X can provide a great amount of energy for liver cells to carry out metabolic reactions. (1)

(c) Rough endoplasmic reticulum (1)
This organelle is responsible for protein synthesis. Liver cells require a lot of enzymes to catalyse metabolic reactions. (1)

(d) Any ***one*** of the following: (1, 1)

- Sperm cell (1)
  This can supply a great amount of energy for the locomotion of the sperm.

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- Muscle cell (1)

  This can supply a great amount of energy for muscle contraction. (1)

(Or other reasonable answers)

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	2
(c)	Understanding basic concepts	1
	Applying concepts	1
(d)	Understanding basic concepts	2

|!|BCESC0030000011|!|

The following lists some types of sub-cellular structures.

A. cell wall
B. cell membrane
C. cytoplasm
D. chloroplast
E. nucleus
F. large vacuole
G. smooth ER
H. mitochondrion

Determine which structures match with each of the descriptions below. Write the appropriate letter in the space provided.

(a) found in plant cells but ***not*** in animal cells (2 marks)

_____

(b) found in prokaryotic cells (2 marks)

_____

(c) bound by a double membrane (2 marks)

_____

##

(a) A, D and F

(All correct: 2; one wrong answer: deduct 1 mark; two or more wrong answers: 0)

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(b) A, B and C
(All correct: 2; one wrong answer: deduct 1 mark; two or more wrong answers: 0)

(c) D, E, and H
(All correct: 2; one wrong answer: deduct 1 mark; two or more wrong answers: 0)

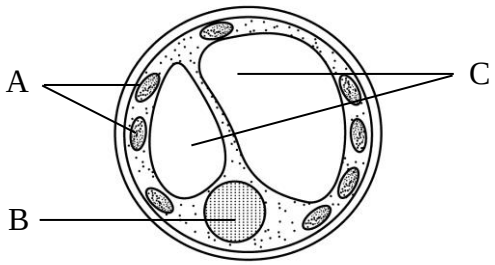
(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	2
(c)	Understanding basic concepts	2

|!|BCESB0030000012|!| ★ **Cross topic**

The diagram below shows a unicellular green alga.



(a) Which kingdom does the green alga belong to? (1 mark)

(b) Name structures B and C. (2 marks)

(c) What gas is produced by structure A when this green alga is exposed to sunlight? (1 mark)

(d) Explain why the green alga does ***not*** need a circulatory system. (1 mark)

##

(a) Kingdom Protista (1)

(b) Structure B: nucleus (1)

Structures C: vacuoles (1)

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(c) Oxygen (1)

(d) The cell is small. Oxygen and nutrients can be distributed efficiently by simple diffusion. (1)

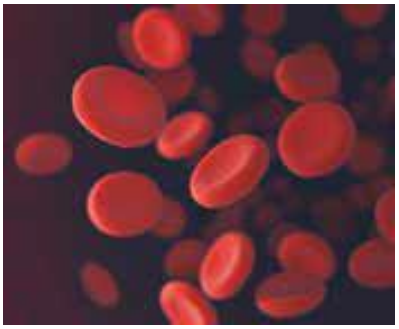
(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Analysing data	2
(c)	Understanding basic concepts	1
(d)	Applying concepts	1

|!|BCESB0030000013|!| ★ **Cross topic**

The photographs below show two types of cells (A and B). A is found in humans, while B is found in a tree.



cell type A



cell type B

(a) Identify cell types A and B. (2 marks)

(b) State ***one*** structural similarity between cell types A and B, which makes them different from a typical cell. (1 mark)

(c) Describe ***one*** structural difference between cell types A and B. (1 mark)

(d) What are the functions of cell types A and B? (2 marks)

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##

(a) A: (mature) red blood cells (1)

B: (mature) xylem cells (1)

(b) Lacking a nucleus (1)

(c) Cell B has cell wall but cell A has no cell wall. / Cell A has cytoplasm but cell B has no cytoplasm. (1)

(d) Cell A: transports oxygen (1)

Cell B: provides support to the plant / forms vessels for transport of water (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	1
(c)	Applying concepts	1
(d)	Understanding basic concepts	2

|!|BCESB0030000014|!| ★

The endoplasmic reticulum (ER) is a network of membrane-bound sacs which are interconnected. There are two types of ER: rough ER and smooth ER.

(a) What is the difference between the structure of rough ER and smooth ER? (1 mark)

(b) How do the rough ER and smooth ER differ in functions? (2 marks)

(c) Explain why human pancreas cells have abundant rough ER. (2 marks)

##

(a) Rough ER has ribosomes attached on its surface while smooth ER has not. (1)

(b) Rough ER helps the synthesis and transport of proteins (1)

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while smooth ER helps the synthesis and transport of lipids. (1)

(c) Pancreas cells require a lot of enzymes for catalysing various metabolic activities / forming hormones for regulating blood glucose level. (1)

Enzymes / hormones are proteins which are synthesized in rough ER. (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)	Applying concepts	2

|!|BCESC0030000015|!|

Complete the following paragraph with suitable words. (5 marks)

Palisade mesophyll cell has a thick outer layer called the (a)_____. This is made up mainly of (b)_____. Its cytoplasm contains many (c)_____ which are involved in the process of photosynthesis. The large central (d)_____ is full of (e)_____ and this helps maintain the shape of the cell.

##

(a) cell wall (1)

(b) cellulose (1)

(c) chloroplasts (1)

(d) vacuole (1)

(e) water (1)

(5 marks)

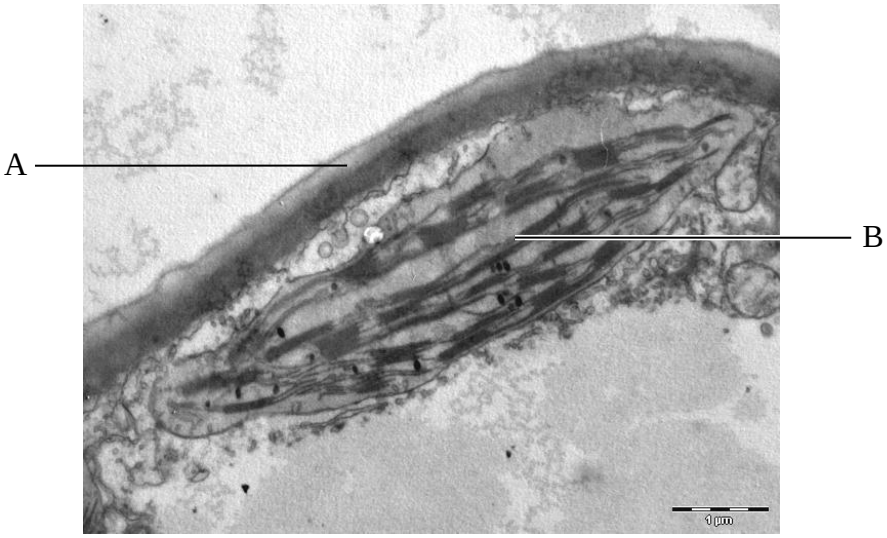
##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

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|!|BCESB0030000016|!| ★

The following micrograph shows part of a plant cell.



(Credits: Andreas Anderluh and Bela Hausmann, Wikimedia Commons, https://commons.wikimedia.org/wiki/File:Chloroplast_12000x_(3660656618).jpg)

(a) Identify cell structures A and B. (2 marks)

_____

_____

(b) State ***one*** function of cell structures A and B. (2 marks)

_____

_____

(c) State ***two*** features that are possessed by this plant cell and cannot be found in a prokaryotic cell. (2 marks)

_____

_____

##

(a) A: cell wall (1)
B: chloroplast (1)

(b) Function of the cell wall: protects the plant cell / gives shape to the plant cell / supports the plant cell (1)
Function of the chloroplast: contains chlorophyll which absorbs light energy for photosynthesis (1)

(c) Any ***two*** of the following: (2)

- It has a true nucleus.

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- It has membrane-bound organelles.
- It has a cell wall made up of cellulose.

(Or other reasonable answers)

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	2
(c)	Applying concepts	2

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